CAPSTONE PROJECT · 2026

Micro-Lending Risk Intelligence System

AI/ML-Based Loan Default Risk Assessment

Presented by: Deesha (2310990067) Group 10 , Semester VII BE-CSE 2023 BATCH

Project: Capstone Project Year: 2026

Technologies

- Python
- Machine Learning
- Scikit-learn
- FastAPI
- Streamlit
- MySQL
- Pandas
- NumPy

FLOW OF PRESENTATION

- INTRODUCTION
- PROBLEM STATEMENT
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- RISK ASSESMENT ENGINE
- LENDING DECISION ENGINE
- STREAMLIT DASHBOARD UI & OUTPUTS & RESULTS
- CONCLUSION RESULTS & FUTURE SCOPE

OVERVIEW

Introduction

1. Our project uses AI and Machine Learning to check borrower details and predict whether they may have difficulty repaying a loan.

2. It looks at important details like loan amount, income, interest rate, DTI, credit history, employment, and loan purpose to calculate a risk score and risk category.

3. The system gives a clear lending recommendation with reasons, helping lenders make faster and more consistent decisions.

4. MySQL, FastAPI, and Streamlit are used to store data, manage the system, and provide an easy-to-use dashboard, making the overall process simple, fast, and easy to understand.

THE CHALLENGE

Problem Statement

1

Micro-lenders often rely on manual verification and traditional credit assessment methods, which are time-consuming, prone to human bias, and may lead to inconsistent lending decisions. This makes it difficult to process loan applications quickly while ensuring fairness and accuracy.

2

With the growing volume of borrower data, identifying hidden risk patterns becomes challenging. Conventional approaches may fail to capture complex relationships in the data and often lack transparency, resulting in inaccurate or uneven risk classification and increased chances of loan defaults.

APPROACH

Proposed Solution

1. Analyze borrower data using AI/ML to understand their financial and credit profile.
2. Predict loan default probability using a trained Machine Learning model.
3. Calculate risk score and category such as Low, Medium, High, or Very High Risk.
4. Provide a lending recommendation with simple reasons for the decision.
5. Use Streamlit, FastAPI, and MySQL to provide an interactive, reliable, and user-friendly system.

GOALS

Objectives

01

To automate borrower risk assessment by using financial and credit information such as income, loan amount, interest rate, credit score, and debt-to-income ratio.

02

To predict and classify loan risk by calculating the probability of default and converting it into an easy-to-understand risk score such as Low, Medium, High, or Very High Risk.

03

To improve lending decisions by providing a clear recommendation (Eligible/Not Eligible) along with the main reasons behind the decision, making the process faster, more consistent, and easier to understand.

Business value

Value to a lender: The platform can reduce manual effort, make screening more consistent and help focus human review on medium- and high-risk applications.

APPROACH

Why This Project Is Different

TRADITIONAL SYSTEM

Borrower Data ↓ Manual Rules ↓ Loan Decision

Static thresholds. Same treatment for every borrower. No visibility into risk drivers.

PROPOSED AI SYSTEM

Borrower Data → Data Engineering → ML Prediction

Default Probability → Risk Scoring → Explainable Decision

Combines machine-learning-based default prediction with an interpretable risk-scoring and lending-decision engine exposed through both API and interactive dashboard interfaces.

MICRO-LENDING RISK INTELLIGENCE

Code pipeline: data becomes a lending decision

Each module has one clear job - analyse, predict, explain and present.

FINAL OUTPUT • Default probability → Risk score → Approve / Manual review / Reject

CODE PIPELINE - GENERATED COMMANDS USED

Task	Command
Check database	python3 -m src.db_check
Generate EDA	python3 -m src.eda
Generate business reports	python3 -m src.business_queries
Train model	python3 -m src.train_model
Run Streamlit interface	python3 -m streamlit run app.py
Run FastAPI backend	python3 -m uvicorn src.api:app --reload

STACK

Technology Stack

DATABASE

- MySQL

BACKEND

- Python
- FastAPI
- Uvicorn

MACHINE LEARNING

- Scikit-learn
- NumPy
- Pandas
- Joblib

FRONTEND / VISUALIZATION

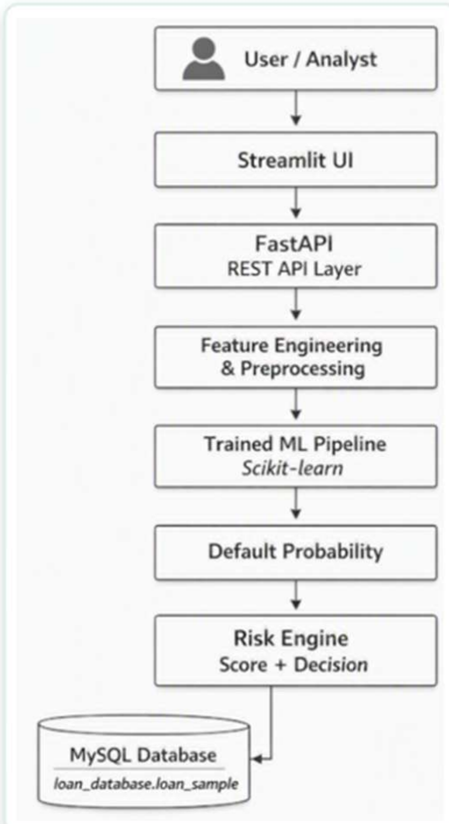
- Streamlit
- Matplotlib / Plotly

DEVELOPMENT

- VS Code
- Python Virtual Env.
- Git / GitHub

SYSTEM ARCHITECTURE

First, the user enters the borrower details through the Streamlit interface. The information goes to FastAPI, which acts as the backend. The data is then cleaned and prepared through feature engineering and preprocessing. After that, the trained machine-learning model predicts the probability of loan default. The risk engine converts this probability into a risk score, risk category, and decision. The MySQL database stores the loan data used by the system.



User / Analyst

Enters the borrower details through the Streamlit interface.

Streamlit UI → FastAPI

The information goes to FastAPI, which acts as the backend.

Feature Engineering & Preprocessing

The data is then cleaned and prepared.

Trained ML Pipeline

Predicts the probability of loan default.

Risk Engine

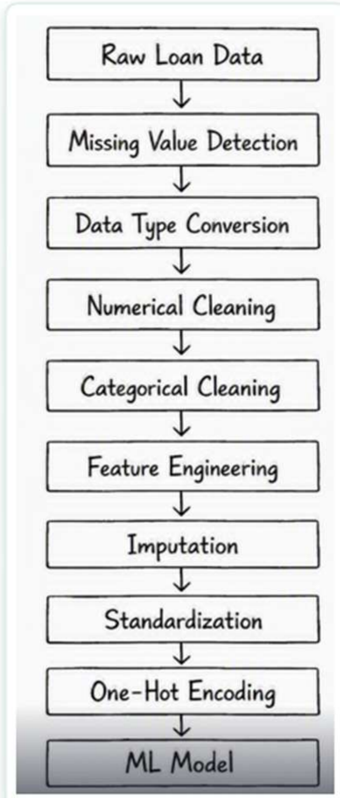
Converts this probability into a risk score, risk category, and decision.

MySQL Database

Stores the loan data used by the system.

DATA PREPROCESSING PIPELINE

The process starts with **raw loan data**. First, missing values and incorrect data types are identified and corrected. Then numerical and categorical data are cleaned. Important features are created, missing values are filled, and the data is standardized. Categorical values are converted into numbers using **one-hot encoding**. Finally, the cleaned data is ready to be given to the **Machine Learning model** for prediction.



Raw Loan Data

The process starts with raw loan data.

Missing Value Detection & Data Type Conversion

Missing values and incorrect data types are identified and corrected.

Numerical & Categorical Cleaning

Numerical and categorical data are cleaned.

Feature Engineering

Important features are created.

Imputation

Missing values are filled.

Standardization

The data is standardized.

One-Hot Encoding

Categorical values are converted into numbers using one-hot encoding.

ML Model

The cleaned data is ready to be given to the Machine Learning model for prediction.

SQL BUSINESS INSIGHTS SCREENSHOT OUTPUT

PORTFOLIO OVERVIEW

Calculates total loans, total loan amount, average loan amount and overall bad-loan percentage. It gives management a quick view of portfolio size, exposure and risk.

```sql
USE loan_database;

-- 1. Portfolio size, exposure and overall default rate
SELECT COUNT(*) AS total_loans,
       ROUND(SUM(loan_amnt), 2) AS total_amount,
       ROUND(AVG(loan_amnt), 2) AS average_loan_amount,
       ROUND(100 * AVG(CASE WHEN LOWER(TRIM(loan_condition)) = 'bad loan
                            THEN 1 ELSE 0 END), 2) AS bad_loan_rate_pct
FROM loan_sample;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_loans	total_amount	average_loan_amount	bad_loan_rate_pct
83373	1264493150.00	15166.70	33.40

SQL BUSINESS INSIGHTS SCREENSHOT OUTPUT

2. Risk by credit grade	**Compares the number of loans, average amount and bad-loan rate for each grade. It helps check whether assigned grades correctly represent increasing credit risk.**

```
-- 2. Which credit grades have the highest bad-loan rate?
SELECT grade, COUNT(*) AS loans, ROUND(AVG(loan_amnt), 2) AS avg_loan_amount,
       ROUND(100 * AVG(CASE WHEN LOWER(TRIM(loan_condition)) = 'bad loan'
                            THEN 1 ELSE 0 END), 2) AS bad_loan_rate_pct
FROM loan_sample GROUP BY grade ORDER BY bad_loan_rate_pct DESC;
```

Result Grid | Filter Rows: | Export:

grade	loans	avg_loan_amount	bad_loan_rate_pct
G	655	20594.05	69.77
F	2148	19283.41	65.41
E	6301	17792.51	57.77
D	13610	15923.83	45.82
C	24716	14953.59	36.66
B	22615	14039.23	24.24
A	13328	14530.26	11.75

SQL BUSINESS INSIGHTS SCREENSHOT OUTPUT

3. Risk by loan purpose	**Shows loan volume, total exposure and bad-loan rate for each purpose. It helps identify high-risk purposes and improve product-level lending rules.**

```sql
-- 3. Which loan purposes create the most risk and exposure?
SELECT purpose, COUNT(*) AS loans, ROUND(SUM(loan_amnt), 2) AS total_amount,
       ROUND(100 * AVG(CASE WHEN LOWER(TRIM(loan_condition)) = 'bad loan'
                       THEN 1 ELSE 0 END), 2) AS bad_loan_rate_pct
FROM loan_sample GROUP BY purpose ORDER BY bad_loan_rate_pct DESC;
```

Result Grid | Filter Rows: | Export: | Wr

purpose	loans	total_amount	bad_loan_rate_pct
educational	23	165050.00	52.17
small_business	1034	17576275.00	46.23
moving	566	4749650.00	40.28
renewable_energy	60	723375.00	36.67
debt_consolidation	47878	767320525.00	35.18
medical	1035	9815425.00	34.88
other	5124	54933575.00	34.29
major_purchase	1822	24478150.00	32.71

SQL BUSINESS INSIGHTS SCREENSHOT OUTPUT

4. Geographic risk	**Calculates bad-loan rates by state while excluding groups with fewer than 25 loans. The minimum count reduces misleading conclusions from very small samples.**

```
-- 4. Geographic risk (minimum 25 loans avoids tiny-sample conclusions)
SELECT addr_state, COUNT(*) AS loans,
       ROUND(100 * AVG(CASE WHEN LOWER(TRIM(loan_condition)) = 'bad loan'
                       THEN 1 ELSE 0 END), 2) AS bad_loan_rate_pct
FROM loan_sample GROUP BY addr_state HAVING COUNT(*) >= 25
ORDER BY bad_loan_rate_pct DESC;
```

Result Grid | Filter Rows:

addr_state	loans	bad_loan_rate_pct
AL	1022	39.92
LA	977	39.00
OK	795	38.24
ND	121	38.02
SD	170	37.65
MS	487	37.58
AK	187	36.90
AR	618	36.57

SQL BUSINESS INSIGHTS SCREENSHOT OUTPUT

5. Interest rate by outcome	**Compares average interest rates for good and bad loans. It shows whether higher-risk borrowers received higher prices and whether pricing aligns with outcomes.**

Result Grid Filter Rows:

loan_condition	loans	average_interest_rate_pct
good loan	55525	12.69
bad loan	27848	15.67

```
-- 5. Does the average interest rate differ for good and bad loans?
SELECT loan_condition, COUNT(*) AS loans,
       ROUND(AVG(CAST(REPLACE(int_rate, '%', '') AS DECIMAL(10,2))), 2)
           AS average_interest_rate_pct
FROM loan_sample GROUP BY loan_condition;
```

SQL BUSINESS INSIGHTS SCREENSHOT OUTPUT

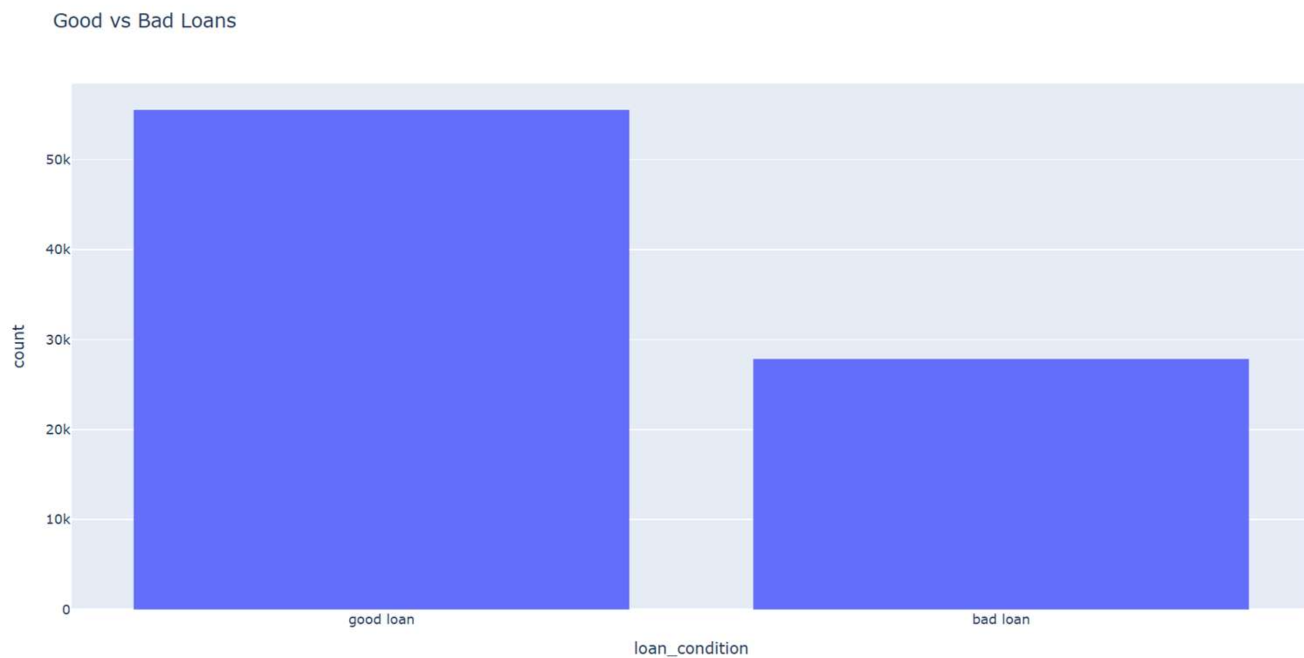
6. DTI risk bands	**Groups borrowers into low, medium, high and very-high DTI bands. It makes the relationship between debt burden and observed default easier to understand.**

```sql
-- 6. Default performance by debt-to-income band
SELECT CASE WHEN dti < 10 THEN 'Low (<10)'
            WHEN dti < 20 THEN 'Medium (10-19.99)'
            WHEN dti < 30 THEN 'High (20-29.99)'
            ELSE 'Very High (30+)' END AS dti_band,
       COUNT(*) AS loans,
       ROUND(100 * AVG(CASE WHEN LOWER(TRIM(loan_condition)) = 'bad loan'
                      THEN 1 ELSE 0 END), 2) AS bad_loan_rate_pct
FROM loan_sample WHERE dti IS NOT NULL GROUP BY dti_band ORDER BY MIN(dti);
```

Result Grid | Filter Rows:

dti_band	loans	bad_loan_rate_pct
Low (<10)	14239	26.78
Medium (10-19.99)	33280	31.14
High (20-29.99)	26067	37.21
Very High (30+)	9787	40.57

EDA OUTPUTS



1. Good vs Bad Loans	**Compares the number of good and bad loans. It helps check class balance and shows whether the model may need special attention to the smaller class.**

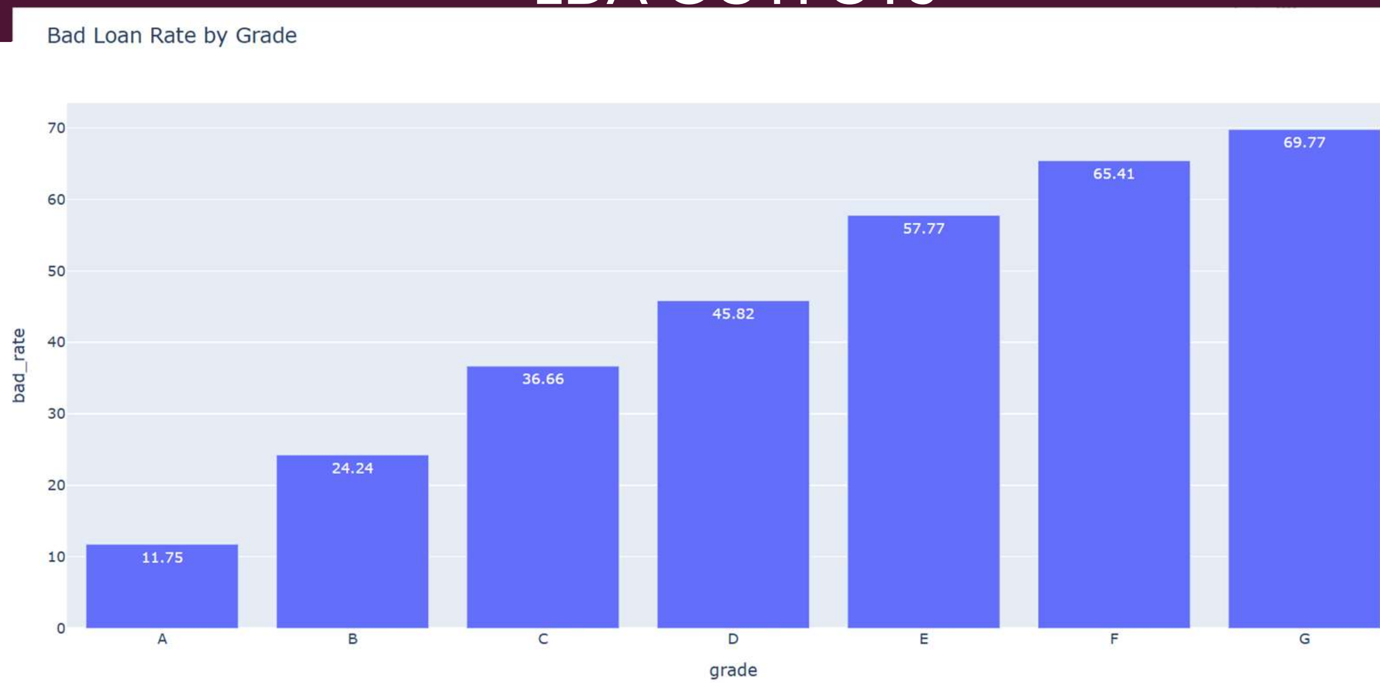
EDA OUTPUTS

2. DTI by Loan Condition	**Compares debt-to-income ratios for good and bad loans. A higher DTI can indicate that a larger share of income is already committed to debt.**

EDA OUTPUTS

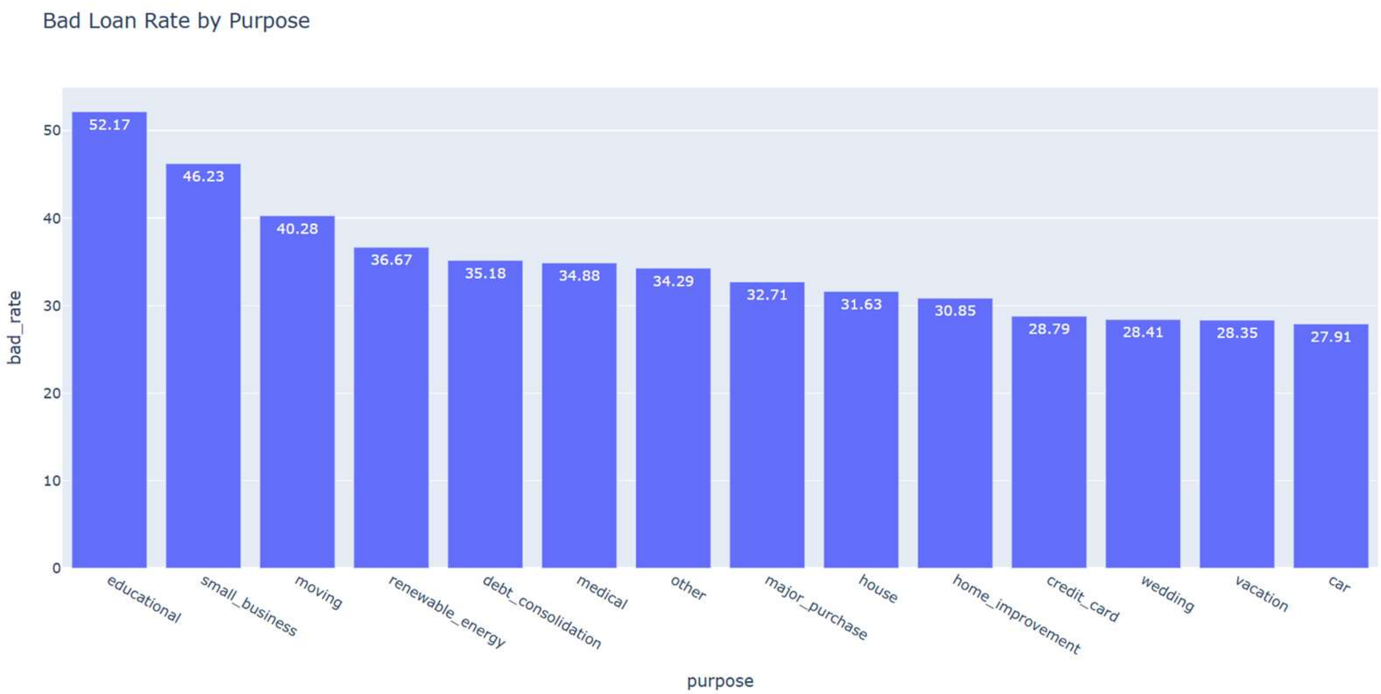
3. Revolving Utilization	**Compares credit utilization between the two loan outcomes. High utilization may show financial pressure and can be associated with greater repayment risk.**

EDA OUTPUTS



4. Bad-Loan Rate by Grade	**Shows the default percentage for each credit grade. It helps verify whether weaker grades carry higher observed risk and supports grade-based lending policy.**

EDA OUTPUTS



5. Bad-Loan Rate by Purpose	**Compares default risk across loan purposes such as debt consolidation, business and credit card. It helps identify purposes that may need tighter review.**

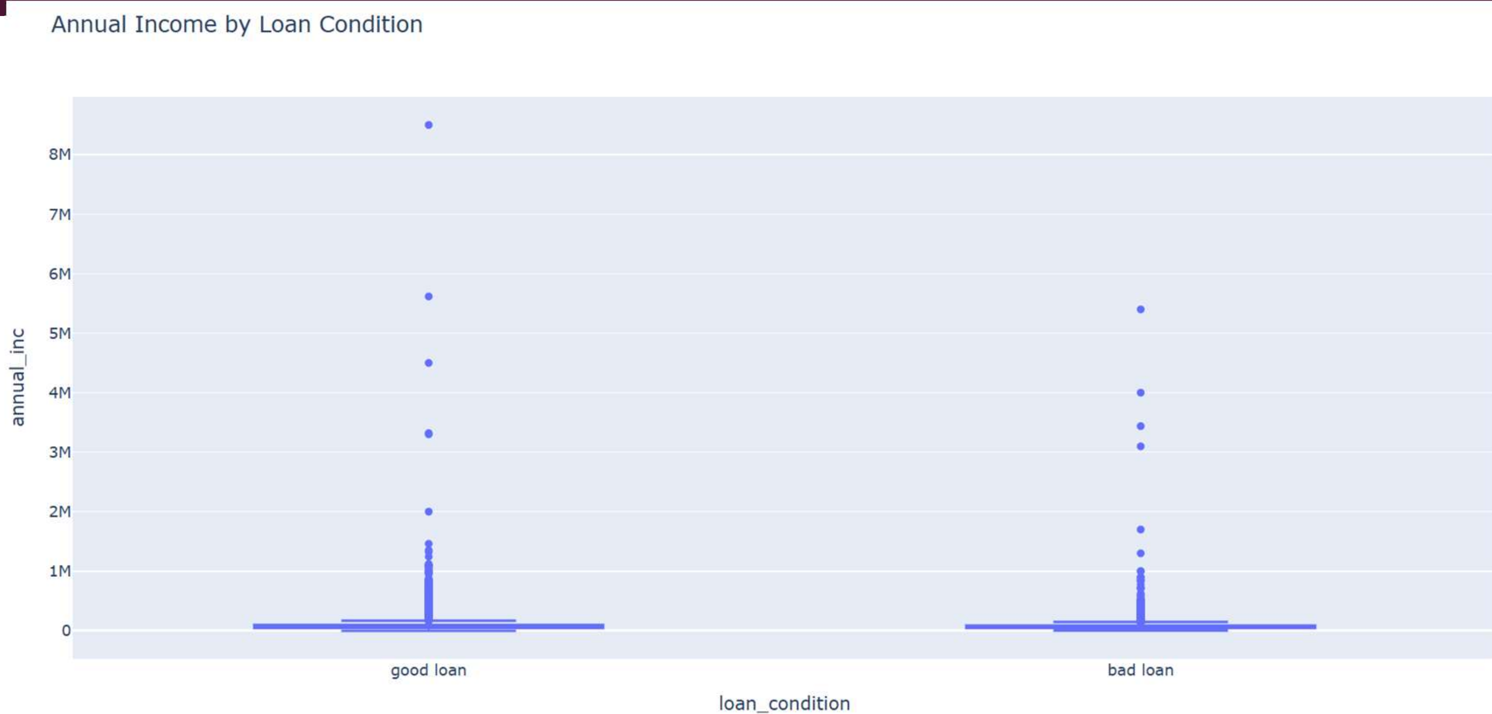
EDA OUTPUTS

6. Loan Amount by Condition	**Compares requested loan amounts for good and bad loans. It helps determine whether larger loans are linked with different repayment outcomes or exposure.**

EDA OUTPUTS

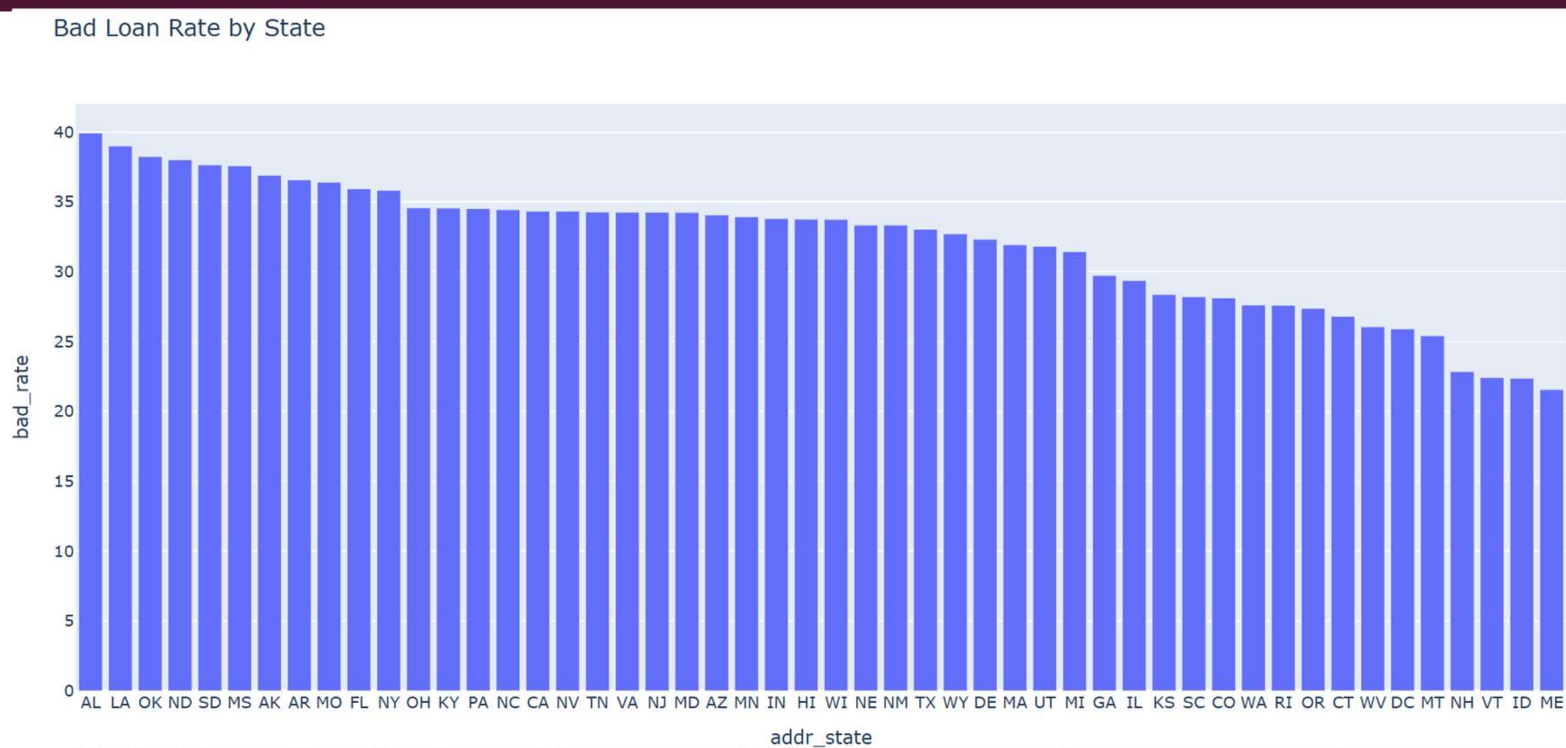
7. Interest Rate by Condition	**Shows how interest rates differ between good and bad loans. Higher rates often represent higher expected risk, but may also increase repayment burden.**

EDA OUTPUTS



8. Annual Income by Condition	**Compares borrower income for good and bad loans. Income is useful for judging repayment capacity, but it should be considered together with loan amount and DTI.**

EDA OUTPUTS



9. Bad-Loan Rate by State	**Compares default rates across borrower states. It supports geographic monitoring and can reveal areas that require further investigation, not automatic discrimination.**

DATA ENGINEERING

Feature Engineering

Feature engineering converts raw financial information into variables that are more useful for machine-learning models.

1 **Loan-to-Income Ratio**

```
Loan-to-Income % = Loan Amount / Annual Income × 100
```

Higher values may indicate greater repayment pressure.

2 **Average FICO**

```
Average FICO = (FICO Low + FICO High) / 2
```

3 **Credit History**

```
Credit History Years = Current Year − Earliest Credit Year
```

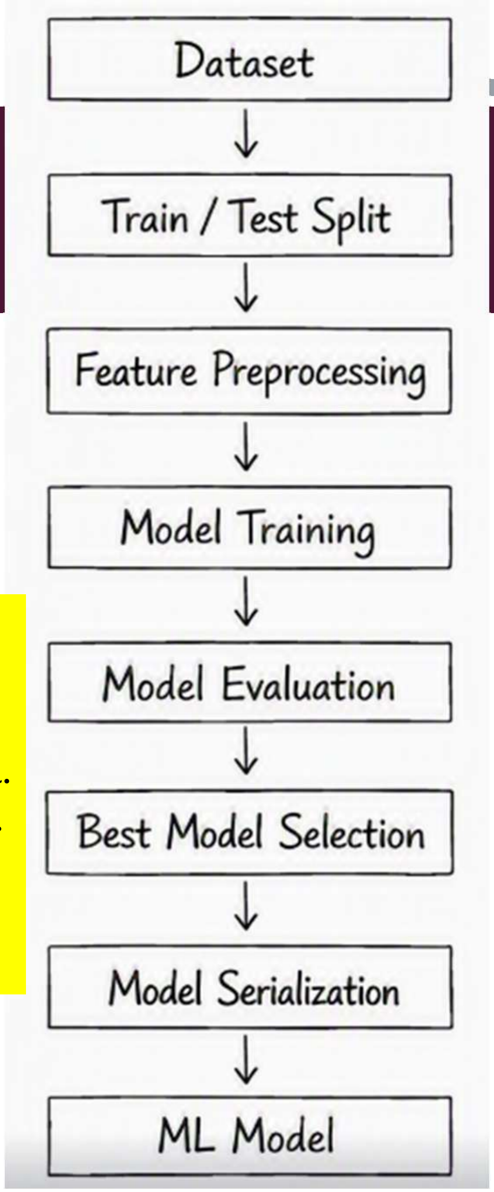
4 **Issue Year**

```
Extracted from the loan issue date.
```

MACHINE LEARNING PIPELINE MODEL DEVELOPMENT

The trained model is stored as:
`models/default_model.joblib`
Metadata is stored as:
`models/model_metadata.json`

- **Dataset:** Collect borrower data containing financial and credit information.
- **Train/Test Split:** Divide the data into training and testing sets.
- **Feature Preprocessing:** Clean and prepare the data for the ML model.
- **Model Training:** Train different machine learning models using the prepared data.
- **Model Evaluation:** Check and compare the performance of the trained models.
- **Best Model Selection:** Select the model that gives the best results.
- **Model Serialization:** Save the selected model for future use.
- **ML Model:** Use the final model to predict the borrower's loan default risk.



RISK INTELLIGENCE

Risk Assessment Engine

This is where we distinguish the project from a simple ML prediction project.

ML OUTPUT

The model produces: Default Probability

For example:

Default Probability = 22%

RISK SCORE

The system converts probability into an intuitive score:

```
Risk Score = (1 − Default Probability) × 100
```

WORKED EXAMPLE

22% default probability

Risk Score = (1 − 0.22) × 100 = 78

A default probability of 22% converts to a Risk Score of 78.

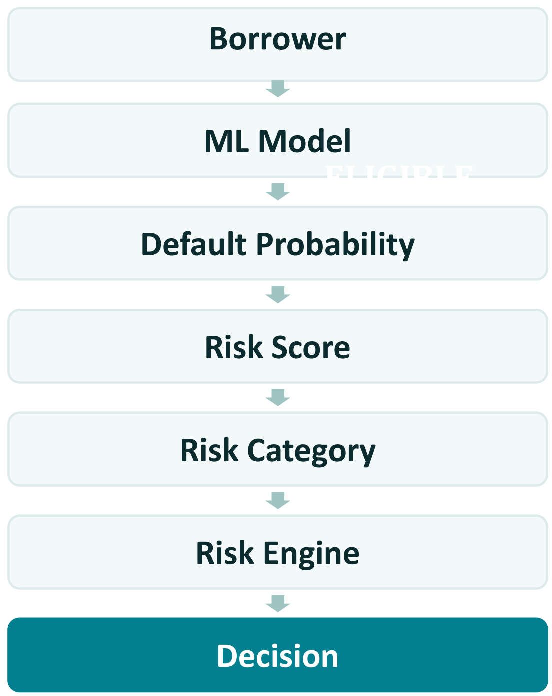
RISK CATEGORIES

75–100	**Low Risk**
55–74	**Medium Risk**
35–54	**High Risk**

RISK INTELLIGENCE

Lending Decision Engine

The system combines the ML result with rule-based risk assessment.



INTEGRATION

FastAPI REST API

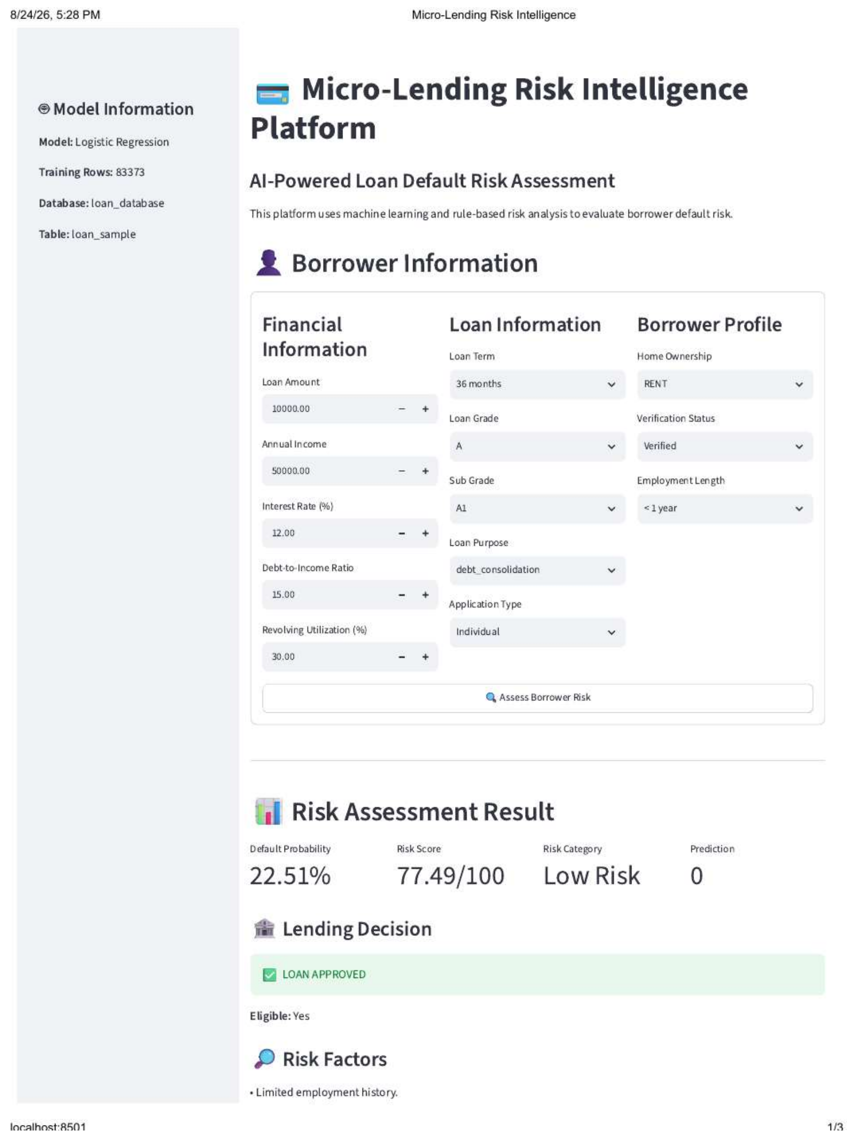
API Endpoints

Endpoint	Purpose
`/`	API information
`/health`	Health check
`/model-info`	Model details
`/predict`	Predict new borrower
`/borrower/{row_number}`	Retrieve borrower
`/predict-existing/{row_number}`	Assess database borrower
`/database-summary`	Database statistics

Why an API layer

FastAPI exposes the trained model and risk engine as REST endpoints — so the Streamlit dashboard, or any other client, can request a borrower assessment or database summary programmatically.

STREAMLIT DASHBOARD



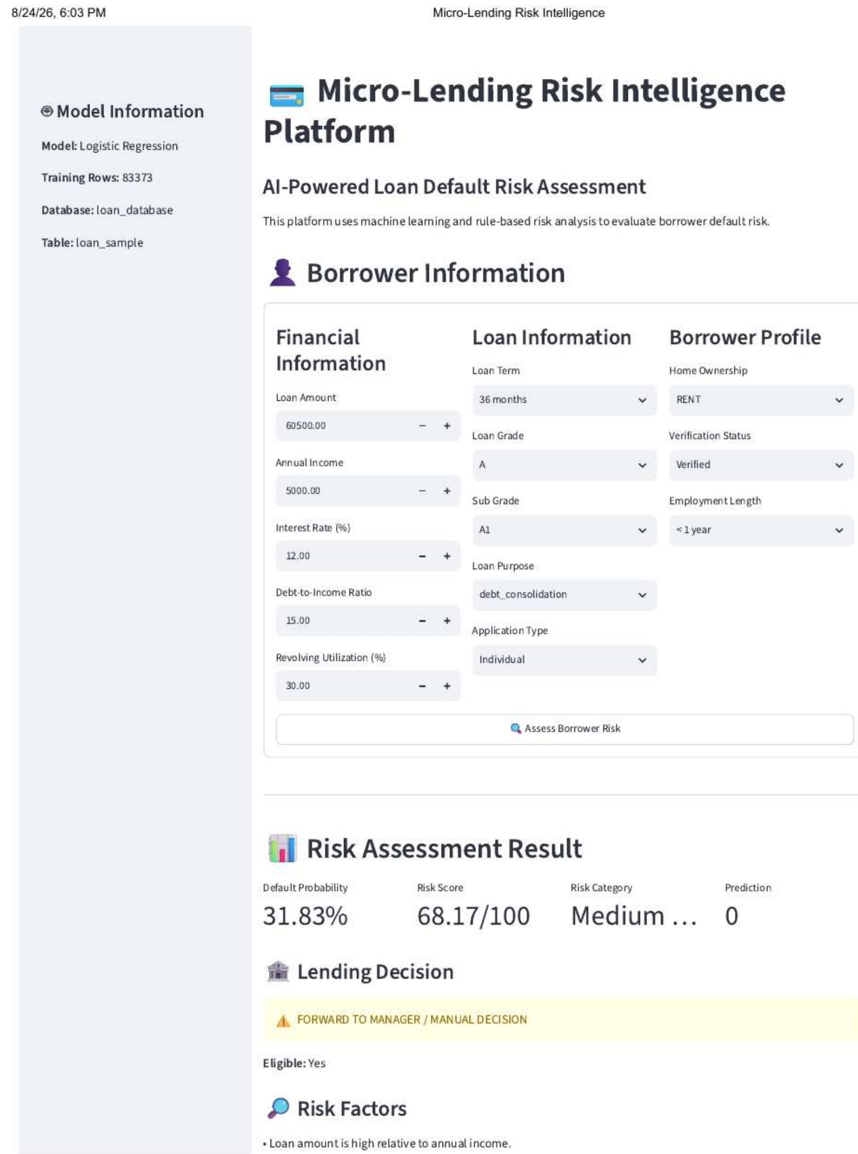
Low-risk approval example

In this report, a loan amount of 10,000 and annual income of 50,000 produced a 22.51% default probability and a risk score of 77.49/100. The borrower was classified as Low Risk and the displayed decision was Loan Approved.

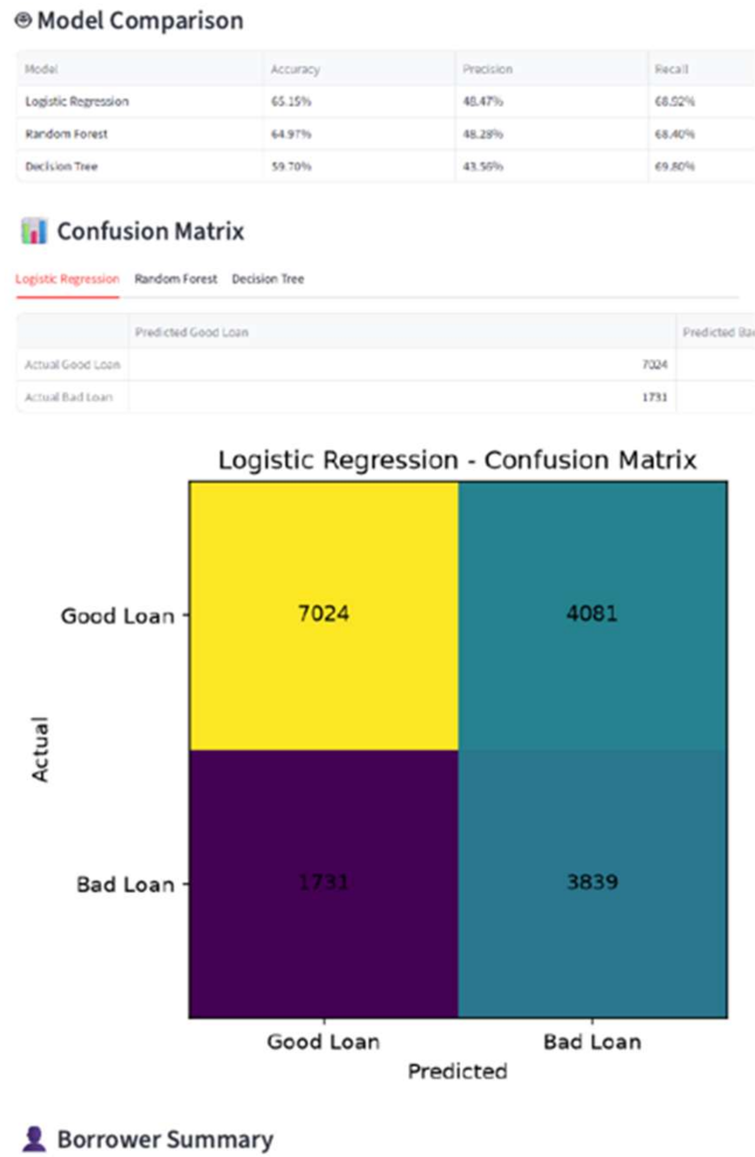
STREAMLIT DASHBOARD

Medium-risk manual-review example

In this report, the loan amount was 60,500 while annual income was 5,000. The system produced a 31.83% default probability, a score of 68.17/100 and a Medium Risk category, so it forwarded the case to a manager for manual decision.



DASHBOARD RESULTS
MODEL COMPARISON AND CONFUSION MATRIX



Model Comparison

Model	Accuracy	Precision	Recall
Logistic Regression	65.15%	48.47%	68.92%
Random Forest	64.97%	48.28%	68.40%
Decision Tree	59.70%	43.56%	69.80%

Confusion Matrix

Logistic Regression Random Forest Decision Tree

	Predicted Good Loan	Predicted Ba
Actual Good Loan	7024	
Actual Bad Loan	1731	

The comparison table shows that Logistic Regression achieved the best overall result among the displayed models. Its recall of about 68.92% means it detected a useful share of actual bad loans, while its ROC-AUC of about 72.06% shows moderate separation between good and bad borrowers.

The Logistic Regression confusion matrix in the attached UI shows 7,024 correctly predicted good loans, 4,081 good loans predicted as bad, 1,731 bad loans predicted as good, and 3,839 correctly predicted bad loans. The 1,731 missed bad loans are important because they represent risky borrowers that may be approved incorrectly.

TARGET AND SELECTED MODEL

The target column is loan_condition: Good Loan is mapped to 0 and Bad Loan is mapped to 1. Logistic Regression was selected because it gave a useful balance of predictive performance and explainability.

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	65.15%	48.47%	68.92%	72.06%
Random Forest	64.97%	48.28%	68.40%	71.82%
Decision Tree*	59.70%	43.56%	69.80%	Shown in UI comparison

Simple interpretation: Recall is especially important in credit risk because it shows how many actual bad loans were detected. ROC-AUC shows the model's overall ability to separate good and bad borrowers.

CONCLUSION

RESULTS

The project successfully integrates MySQL + ML + Risk Engine + FastAPI + Streamlit into a single loan-risk intelligence platform.

KEY OUTCOMES

Risk Assessment
Automatically checks the borrower's risk.

Default Prediction
Predicts the chance of loan default.

Loan Recommendation
Suggests suitable lending decisions.

Dashboard & API
Provides an easy dashboard and API for use.

FUTURE SCOPE

Explainable AI

Show why the loan was approved or rejected.

Fraud & Fairness

Detect fraud and reduce unfair decisions.

Cloud & Recommendations

Run the system online and suggest a suitable loan amount.

Thank you.